VEPHLA UNIVERSITY

School of Data Sciences and Data Engineering

CAPSTONE PERSONAL PROJECT

EXPLORING DATA

Web Scraping, Data Analysis, and Dashboard Creation

PROJECT TITLE

BookScope

A Data Driven Exploration of Book Prices, Ratings and Categories

SUBMITTED BY

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1. Executive Summary

This report documents the individual Data Analytics Capstone Project completed by Ayodele Modupe (Student ID: VEPH/20C/DA127) for the School of Data Sciences and Data Engineering, Vephla Uni. The formal project title is “Exploring Data: Web Scraping, Data Analysis, and Dashboard Creation.” BookScope is the name used for the dashboard.

The project follows an end-to-end analytics workflow: selecting a public website, collecting catalogue data with Python Requests and BeautifulSoup, preparing the records with pandas, carrying out exploratory data analysis (EDA), creating multiple visualizations, and presenting selected findings in an interactive Power BI dashboard.

The scraper collected 1,000 book listings from 50 catalogue pages. The final enriched dataset has 1,000 rows and five columns: title, price, rating, availability, and category. Price text was converted to numeric values, star-rating words were mapped to integers from 1 to 5, duplicate rows were checked, and category data was merged from book detail pages. There are 67 missing category values because invalid page text was excluded rather than treated as a genuine category.

The mean listed price was £35.07 and the median was £35.98. Prices ranged from £10.00 to £59.99. The mean rating was approximately 2.92 out of 5. The price-rating correlation was approximately 0.028, suggesting almost no linear relationship in this dataset. The interquartile-range method did not flag any price outliers.

The BookScope dashboard includes KPI cards, category/rating/availability slicers, and visuals for rating counts, rating shares, price comparisons, category counts, category distribution, and price distribution. The source is a practice website, so its values should not be interpreted as real-world retail prices or genuine customer ratings.

2. Project Background and Institutional Context

This project was completed for the Vephla Uni. School of Data Sciences and Data Engineering, Data Analytics Capstone Project (LSD – Option 2), for the award of a Nanodegree in Data Analytics. The assignment requires practical demonstration of web scraping, data cleaning, exploratory analysis, visualization, dashboard design, and written communication.

Python techniques and pandas were the preferred tools specified in the brief. Python was used for scraping, data preparation, analysis, and chart generation, while Power BI Desktop was used to build the interactive dashboard. This combination separates analytical work from the final presentation layer.

The formal project title remains “Exploring Data: Web Scraping, Data Analysis, and Dashboard Creation.” BookScope is the project/dashboard brand, with the subtitle “A Data-Driven Exploration of Book Prices, Ratings and Categories.”

The project brief requires at least 1,000 data points, a cleaned CSV, a Python script, at least seven different visualization types, an interactive dashboard, and a report in PDF format. The submission instructions

additionally request an extensive report of at least 15 pages and a recorded presentation of approximately 15 minutes.

3. Project Objectives and Scope

3.1 Main objective

The main objective was to demonstrate an end-to-end data analytics process by collecting book catalogue data from a public website, preparing and analysing the data, and presenting the results in a user-friendly interactive dashboard.

3.2 Specific objectives

- Use Requests to retrieve catalogue web pages and BeautifulSoup to parse their HTML.
- Collect at least 1,000 book records and preserve the raw extraction.
- Use pandas to inspect, clean, standardize, enrich, and export the dataset.
- Calculate descriptive statistics, grouped summaries, correlation, and an outlier check.
- Create more than seven types of visualizations to explore different aspects of the data.
- Build a Power BI dashboard with KPI cards, slicers, and multiple charts.
- Document methods, findings, limitations, and recommendations.

3.3 Scope boundaries

The project analyses the catalogue information collected from Books to Scrape. It does not measure the real publishing industry, actual commercial book prices, genuine customer satisfaction, or live retail availability. The website is designed for scraping practice; its displayed prices and ratings are treated as demonstration-site values.

4. Website Selection and Dataset Description

The selected source was Books to Scrape (https://books.toscrape.com/), a public practice website for learning web scraping. It contains a catalogue of books with repeated listing structures and multiple pages. Each listing provides a title, price, star rating, and availability status. Book detail pages provide category information.

The source was suitable because its consistent structure made it possible to automate extraction, it provided enough records to meet the 1,000-record requirement, and its fields supported a range of useful analyses. Automating the collection also reduced manual copying and made the process repeatable.

Field	Meaning	Preparation / use

title	Book name	Retained as text and used for identification/counting.
price	Listed price	Currency symbol and unwanted text removed; converted to numeric.
rating	Star rating	Words One–Five mapped to integers 1–5.
availability	Stock-status text	Retained for filtering and grouping.
category	Book category	Collected from detail pages and merged; unavailable values remain missing.

The initial catalogue scrape contained four fields: title, price, rating, and availability. Category was collected in a second step and merged into the main dataset by title. The final enriched dataset contains 1,000 rows and five columns.

5. Tools and Technical Environment

5.1 Python and Jupyter Notebook

Python was used as the primary programming language. The notebook workflow allowed the work to be completed in stages, with intermediate outputs inspected before continuing. The notebook was also exported as a Python script for submission and review.

5.2 Requests

Requests sent HTTP requests to the catalogue pages and retrieved their HTML content. The script iterated through 50 catalogue pages and processed each page using the same extraction logic.

5.3 BeautifulSoup

BeautifulSoup parsed the HTML returned by Requests. It located repeated book listing elements and extracted the title, price, rating class, and availability text. A second scraping step visited book detail pages to retrieve category information.

5.4 pandas

pandas stored the extracted records in DataFrames and supported inspection, text cleaning, type conversion, duplicate checks, missing-value assessment, merging category information, group-by calculations, descriptive statistics, and CSV export.

5.5 Visualization tools

Python plotting tools were used to produce exploratory charts saved as image files. Power BI Desktop was used to build the BookScope dashboard with summary cards, filters, and charts. This workflow allowed analysis in Python and interactive exploration in Power BI.

5.6 Project organization

The project was organized into separate folders for raw data, cleaned data, Python work, EDA, visualizations, dashboard, report, presentation, and final submission. The raw file was preserved separately from the cleaned CSV so that cleaning decisions could be reviewed or repeated.

6. Web-Scraping Methodology

6.1 Catalogue-page collection

The scraper iterated through 50 catalogue pages, with approximately 20 books per page, targeting 1,000 records. For each page, Requests retrieved the HTML and BeautifulSoup parsed the page. The script located each book listing and extracted the required attributes.

6.2 Extracting listing information

The title was obtained from the book listing or linked title element. Price was collected as displayed text. Rating was read from the HTML class representing the star label, and availability was extracted from the stock-status text. The values were assembled into structured records and converted to a pandas DataFrame.

6.3 Category enrichment

Category information was collected separately by visiting book detail pages and reading the category from the page navigation/breadcrumb text. The category records were merged into the main catalogue by title. The category table contained 999 unique titles because “The Star-Touched Queen” appeared twice in the catalogue; both listings were associated with Fantasy. The merge preserved all 1,000 catalogue rows.

6.4 Automation and responsible use

Automation made the process more consistent and reduced manual effort. The workflow can be repeated if the website’s structure remains compatible with the selectors used in the code. For other websites, a scraper should respect the site’s terms and access rules, use reasonable request rates, and avoid collecting personal or restricted information.

7. Raw Data Storage and Initial Inspection

The full catalogue extraction was saved as a raw CSV file in the raw-data folder. A first-page test file was also created during development, but it was not combined with the 1,000-row scrape because that would duplicate the first 20 listings.

Initial inspection examined the DataFrame shape, field names, sample records, data types, missing values, and duplicate rows. These checks established whether the scraper had collected the expected number of records and whether the fields were usable for analysis.

The website presents values for people to read, so price and rating initially appeared as text rather than analysis-ready numbers. Price values included currency symbols and an encoding artefact, while rating labels were words. Cleaning these fields was necessary before numerical summaries and charts could be trusted.

The initial cleaned core dataset contained 1,000 rows and four columns, with no missing values in those four fields and no duplicate full rows detected in the first check. Category enrichment was then assessed separately.

8. Data Cleaning and Standardization

8.1 Price conversion

The price field was cleaned by removing the displayed currency symbol and unwanted encoding text, including the stray “Â” character. The cleaned values were converted to numeric data so pandas could calculate averages, medians, minimums, maximums, grouped prices, and correlations, and so Power BI could use price as a numeric axis.

8.2 Rating conversion

The website’s rating labels were represented by words such as One, Two, Three, Four, and Five in the HTML class information. These were mapped to integers 1, 2, 3, 4, and 5. This preserved the rating order and enabled frequency counts, averages, and correlation calculations.

8.3 Text fields and consistency

Book titles were retained as text. Availability was kept as a text field for grouping and dashboard filtering. Valid category names were preserved as labels. The data types were chosen according to the role of each field.

8.4 Duplicate checks

The DataFrame was checked for duplicate full rows. No exact duplicate rows were found in the initial core dataset. A repeated title was observed during category enrichment. Because the goal was to preserve the 1,000 collected catalogue records, repeated listing rows were not automatically deleted solely because their titles matched.

8.5 Missing and invalid categories

Some detail-page extraction results returned “Add a comment” where a category was expected. This was not a valid category label, so it was replaced with a missing value rather than retained as misleading data. The final enriched dataset contains 67 missing category values. These were not filled with guessed categories.

8.6 Final export

The final DataFrame was exported as `02_Clean_Data/books_cleaned.csv`. It contains 1,000 rows and five columns: title, price, rating, availability, and category. This file was used for EDA and imported into Power BI.

9. Data Quality Assessment and Limitations

Data quality checks focused on completeness, consistency, uniqueness, and analytical suitability. The final row count met the 1,000-record requirement. Price and rating were converted to numeric types, and invalid category text was excluded from the valid category labels.

Quality check	Result	Interpretation
Record count	1,000 rows	Meets the minimum requirement.
Final fields	5 columns	Title, price, rating, availability, category.
Core-field missingness	0 in initial cleaned core data	The four original catalogue fields were complete in the initial cleaned pass.
Category missingness	67 values	Category extraction was incomplete; values remain missing.
Duplicate rows	0 exact duplicate full rows in initial core check	No exact duplicates were detected by that check.
Price	Numeric	Suitable for calculations and numeric chart axes.
Rating	Integer 1–5	Suitable for rating frequencies and ordered comparisons.

The 67 missing category values are the main known completeness limitation. Category totals and averages should therefore be interpreted with care. A future improvement would be to inspect the affected detail pages, use a more robust breadcrumb selector, and merge on a stable product URL or product identifier rather than title alone.

The source is a demonstration website, not a live commercial retailer. The prices and ratings should not be generalized to real-world book markets. A title is also not a perfect unique identifier: different products can share titles, and repeated listings can occur. Retaining product URLs would strengthen future merges.

10. Exploratory Data Analysis: Summary Statistics

EDA was performed after the numeric fields and category values were prepared. The purpose was to understand the overall distribution of prices and ratings before interpreting charts or designing the dashboard.

Metric	Result	Meaning
Number of books	1,000	Total catalogue records analysed.
Mean price	£35.07	Average listed price.
Median price	£35.98	Middle price in sorted order.
Minimum price	£10.00	Lowest listed price observed.
Maximum price	£59.99	Highest listed price observed.
Standard deviation	£14.45	Shows spread around the mean.

First quartile (Q1)	£22.11	Approximately 25% of values are at or below this point.
Third quartile (Q3)	£47.46	Approximately 75% of values are at or below this point.
Mean rating	2.92 / 5	Average rating is just below the midpoint.
Median rating	3	Middle rating is three stars.
Mode rating	1	One star was the most frequent rating.

The mean and median prices were relatively close, with the median slightly higher than the mean. Prices nevertheless covered a broad range, from £10.00 to £59.99. The standard deviation of £14.45 indicates meaningful variation in listed prices.

Rating counts were spread across the five possible values: one star appeared 226 times, two stars 196 times, three stars 203 times, four stars 179 times, and five stars 196 times. One star was the largest group, while four stars was the smallest.

11. EDA: Grouping, Correlation and Outliers

11.1 Average price by rating

The mean prices by rating were: one star £34.56; two stars £34.81; three stars £34.69; four stars £36.09; and five stars £35.37. The four-star group had the highest average price, but the differences were small relative to the full price range. This does not prove that higher-priced books receive better ratings.

11.2 Correlation between price and rating

The correlation between price and rating was approximately 0.028. This is very close to zero and indicates almost no linear relationship between these two variables in this dataset. Price alone does not appear to explain the rating values in a linear way. Correlation does not establish causation, and this result applies only to the collected demonstration data.

11.3 Category summaries

Grouped analysis compared record counts and average price/rating across categories. The leading categories by count included Default (152 books), Nonfiction (110), Sequential Art (75), Fiction (65), Young Adult (54), Fantasy (48), Romance (35), Mystery (32), Food and Drink (30), and Children's (29). Default is a category label present in the source and was not treated as invalid.

Among the categories in the top-category summary, Fantasy had the highest average price at £39.59. Young Adult had an average rating of 3.30, while Fiction averaged 3.18. Because 67 records have missing categories, category-level summaries should be read with that limitation in mind.

11.4 Outlier check

The interquartile range method was used to check for extreme prices. Q1 was £22.11 and Q3 was £47.46, giving an IQR of £25.35. Applying the 1.5 × IQR rule gave a lower bound of approximately -£15.92 and an upper bound of approximately £85.48. No observed prices fell outside these bounds, so the rule flagged zero price outliers. The negative lower bound is a mathematical result of the method, not a possible retail price.

Not finding IQR outliers does not prove that every record is correct. It only means that no price fell outside the selected statistical thresholds. The invalid category text was identified through data-quality checks rather than price outlier detection.

12. Python Visualizations

Python charts were created during EDA and saved as image files in the project's visualization folder. The charts examine counts, distributions, rankings, group comparisons, and relationships. More than seven visualization types were created, meeting the assignment requirement.

Visualization	Purpose
Bar chart: book ratings	Compares counts across ratings 1–5.
Bar chart: average price by rating	Compares mean price within each rating group.
Histogram: book price distribution	Shows frequency across price intervals.
Bar chart: top 10 expensive books	Highlights the highest listed prices.
Bar chart: top 10 cheapest books	Highlights the lowest listed prices.
Scatter plot: price vs rating	Explores the relationship between two numeric fields.
Pie chart: rating share	Shows each rating group's share of the catalogue.
Box plot: price by rating	Compares price spread across rating groups.
Heatmap: price-rating relationship	Visually summarizes numeric relationship/correlation.
Bar chart: average price by top categories	Compares average price across selected categories.
Line chart: average price by rating	Shows grouped mean price across the ordered rating scale.

Each chart answers a different question. Bar charts compare counts and group averages; the histogram and box plot describe distributions; the scatter plot explores a relationship; the pie chart displays shares; and the heatmap provides a compact view of numeric relationships. The combination makes the findings easier to communicate than a table alone.

13. BookScope Interactive Dashboard

Power BI Desktop was used to create BookScope from the cleaned CSV. The report is arranged as a single dashboard page so users can view the main indicators and visuals together. A navy sidebar, blue and teal accents, a light canvas, and clear titles create a consistent visual identity.

13.1 KPI cards

The dashboard includes cards for Total Books, Average Price, Average Rating, and Price Range. The expected values from the EDA are 1,000 books, mean price £35.07, average rating approximately 2.92/5, and a price range from £10.00 to £59.99. The cards provide a quick overview before the user explores the charts.

13.2 Slicers and interactivity

The filter area includes slicers for category, rating, and availability. Users can select a category or rating, or filter by availability. The visuals and summary values are intended to update in response to those selections, allowing exploration of subsets without editing the source CSV.

13.3 Dashboard charts

- Number of Books by Star Rating: compares book counts across all five rating levels.
- Share of Books by Rating: a donut chart showing each rating group's share.
- Average Book Price by Star Rating: compares average prices across rating levels.
- Price vs Rating: a scatter plot intended to show price-rating observations.
- Top Categories by Number of Books: a bar chart comparing selected leading categories.
- Book Distribution by Category: a treemap showing relative sizes of selected categories.
- Book Price Distribution: a column chart using price bins to show counts within price intervals.

The KPI cards provide context, slicers narrow the dataset, and the charts offer complementary views of price, rating, and category patterns.

14. Dashboard Interpretation and Quality Checks

The dashboard brings together the key patterns from EDA. The rating charts compare the five rating groups, the price visuals describe the range and distribution of listed prices, and the category visuals highlight differences in category size. The filters make it possible to explore subsets of the catalogue.

Before final submission, the dashboard should be checked for consistency between labels, measures, and units. In the supplied screenshot, some visuals display the euro symbol (€), although the source dataset uses pound sterling (£). The currency format should be changed to pounds throughout.

The screenshot also shows a visual titled "Average Book Price by Star Rating" with an axis labelled as a sum of price and values in the thousands. This suggests the field may be summarized as Sum instead of Average. To

match the title, the price field should be set to Average; the values should then be around the £30–£40 range rather than totals in the thousands.

The scatter plot should be checked to ensure it represents the intended individual records rather than aggregated values. Price should be a numeric X-axis, rating a numeric Y-axis, and title can be used for identification or tooltips. Some overlap is expected because rating has only five possible values.

The price distribution chart should use `price (bins)` on the X-axis and Count of `title` on the Y-axis. If the axis displays individual prices rather than intervals, it may be using the original price field instead of the bin field. These checks are important because visual design must be matched by accurate measures.15. Key Findings and Discussion

15.1 Prices vary across the catalogue

The mean price was £35.07, the median was £35.98, and the observed range was £10.00–£59.99. The histogram and box plot help show this spread, while the top-ten charts identify the most and least expensive records.

15.2 Ratings are spread across five levels

All five rating levels were represented. One-star ratings were most frequent at 226 books, while four-star ratings were least frequent at 179. The mean rating was approximately 2.92/5, close to three stars. The bar and donut charts communicate this distribution clearly.

15.3 Price and rating have little linear association

The correlation of about 0.028 is very small. The results therefore do not support a claim that higher prices are meaningfully associated with higher ratings in a linear way. The scatter plot provides a visual complement to the calculation.

15.4 Group differences are modest

Mean prices across rating groups ranged from £34.56 to £36.09. The four-star group had the highest average, but the difference is modest and should not be overstated. The dataset does not contain other factors that might influence both price and rating.

15.5 Category coverage is incomplete

Category counts and averages show differences across labels, but 67 records have no valid category. Category-based conclusions should therefore be treated as descriptive rather than complete representations of all records.

16. Challenges, Lessons Learned, and Recommendations

16.1 Challenges encountered

- Collecting records across multiple pages required a loop and consistent HTML selectors.
- Prices and ratings were displayed as human-readable text and needed conversion.
- Category extraction required a second detail-page scrape and a merge.
- Some pages returned “Add a comment” where a category was expected; these values were converted to missing rather than treated as real categories.
- A repeated title meant catalogue rows could not safely be deleted solely because titles matched.
- Power BI visuals required careful checking of field wells, summarization, filters, axis labels, and currency formats.

16.2 Lessons learned

The project shows that scraping is only the first step of an analytics workflow. Raw HTML values must be cleaned before calculations can be trusted, and the choices made during cleaning affect every subsequent chart. Preserving the raw dataset also makes the process more transparent and allows cleaning logic to be revisited.

16.3 Recommendations

- Improve category scraping with a more reliable breadcrumb selector and validate extracted values against the page structure.
- Retain product URLs or another stable identifier to make merges more reliable than title-only matching.
- Record the scraping date and add error handling so failed requests can be identified and retried.
- Automate validation checks for row count, allowed rating values, numeric prices, duplicates, and missing values.
- Change dashboard currency labels to GBP (£) and confirm that average-price visuals use Average rather than Sum.
- Test each slicer and confirm that KPI cards and charts respond correctly.
- Keep the notebook, exported Python script, cleaned CSV, Power BI file, report, and presentation link together in the final submission folder.

17. Conclusion

This individual capstone applied the main stages of data analytics to a public book catalogue. Requests and BeautifulSoup were used to collect 1,000 records from 50 catalogue pages. pandas was used to clean the values, convert price and rating fields, check duplicates, enrich the data with categories, and export a final 1,000-row, five-column CSV.

EDA calculated descriptive statistics, rating counts, grouped average prices, a price-rating correlation, category summaries, and an IQR-based outlier check. The mean price was £35.07, the median was £35.98, the mean rating was about 2.92/5, and price-rating correlation was about 0.028. No prices were flagged as outliers by the IQR rule. The main data limitation was 67 missing category values.

More than seven types of Python visualizations were created, and selected findings were presented in the BookScope Power BI dashboard. KPI cards and slicers make the report easier to explore, while the charts show rating frequencies, category distributions, price distributions, and the relationship between price and rating.

The project meets the central aims of web scraping, pandas-based cleaning, EDA, multiple visualizations, and interactive dashboard creation. Before submission, the dashboard should be quality-checked for currency symbols, aggregation settings, and correct use of price bins. Future work should improve category extraction, use stable product identifiers, and strengthen automated validation.

Because Books to Scrape is a demonstration website, the results are best understood as a practical exercise in data analytics rather than evidence about the real book market.

Appendix A. Deliverables and File Checklist

Use this checklist to prepare the final submission and confirm that each file is saved in its intended folder.

- Python notebook: `03_Python/Web_Scraping_Project.ipynb`.
- Exported Python script: `03_Python/Web_Scraping_Project.py`.
- Raw catalogue data: `01_Raw_Data/books_raw.csv`.
- Cleaned dataset: `02_Clean_Data/books_cleaned.csv`.
- Python chart images: `05_Visualizations/`.
- Final Power BI report file: save the `.pbix` file in `06_Dashboard/`.
- Project report: PDF and editable DOCX in `07_Report/`.
- Recorded presentation of approximately 15 minutes and its shareable link.
- Final submission folder and links to the documents/scripts.

The brief also requests the student’s full name, a high-quality headshot, links to project files and scripts, a presentation link, and an updated portfolio link. The dashboard file itself should be included or shared using the submission system’s accepted method.

Appendix B. Reproducibility and Interpretation Notes

The notebook is the most useful record of the workflow because it preserves the sequence of scraping, cleaning, EDA, and chart creation. The exported Python script provides a linear version of the code for review. The cleaned CSV is the common data source for the analysis and Power BI dashboard.

To reproduce the analysis, run the notebook in order from the imports and scraping cells through cleaning, category enrichment, EDA, chart export, and final CSV export. The website's HTML structure may change, so selectors may need adjustment if scraping is repeated later. The numerical results in this report refer to the dataset collected and prepared for this project.

Interpret the results descriptively. The small price-rating correlation indicates little linear association in this dataset; it does not prove price can never affect ratings. The absence of IQR outliers means the selected statistical rule did not flag extreme prices, not that every record is error-free. Category averages should be considered alongside the 67 missing category values.

The final presentation should explain the project purpose, demonstrate the website and scraping approach, describe cleaning decisions, show the EDA and visualization types, demonstrate dashboard filters and charts, discuss findings and limitations, and conclude with recommendations. Keep the presentation aligned with the actual notebook, cleaned data, and final dashboard.